

HOT- WORKING PROCESSES

- Shaping of metal by deformation is a very very old tradition.
- Processes such as rolling, wire drawing etc. were common in the Middle Ages. In North America, by 1680 the Saugus Iron Works near Boston had an operating drop forge, rolling mill, and slitting mill.
- Although basic concepts of many forming processes have remained largely unchanged throughout history, details and equipment have evolved considerably.

INTRODUCTION

- Manual processes were converted into machine processes during the industrial revolution.
- The machinery then became bigger, faster, and more powerful.
- Waterwheel power was replaced by steam and then electricity.
- More recently, computer-controlled, automated operations have emerged

CLASSIFICATION OF DEFORMATION PROCESSES

- Processes divide into the following :-
 - Primary processes reduce a cast material into intermediate shapes, such as slabs, plates, or billets.
 - Secondary processes further convert these shapes into finished or semi-finished products.

CLASSIFICATION OF DEFORMATION PROCESSES

- Bulk deformation processes

- Surface area of work piece changes significantly.
- Thicknesses or cross sections of material are reduced
- Shapes are changed.
- As volume of material remains constant, other dimensions must change in proportion.
- Enveloping surface area is altered, usually increasing as product lengthens.

CLASSIFICATION OF DEFORMATION PROCESSES

- Sheet-forming operations

- Involve deformation of material where thickness and surface area remain relatively constant.
- Coining, for example, begins with sheet material but alters the thickness in a complex manner that is essentially bulk deformation.

HOT-WORKING PROCESSES

- Hot-working processes provide means of producing a desired shape.
- At elevated temperatures, metals weaken and become more ductile.
- With continual re-crystallization, massive deformation take place without exhausting material plasticity.

HOT-WORKING PROCESSES

- Some major modern hot-working manufacturing processes are:
 - Rolling
 - Forging
 - Extrusion
 - Hot drawing
 - Pipe welding
 - Piercing

ROLLING

- Rolling is usually first process used to convert material into a finished wrought product.
- Stock can be rolled into blooms, billets, slabs, or these shapes can be obtained directly from continuous casting.
 - A bloom has a square or rectangular cross section, with a thickness greater than 6 inches and a width no greater than twice the thickness.

ROLLING

- A billet is usually smaller than a bloom and has a square or circular cross section. Billets are usually produced by some form of deformation process, such as rolling or extrusion.
- A slab is a rectangular solid where the width is greater than twice the thickness. Slabs can be further rolled to produce *plate, sheet, and strip*

ROLLING

- These hot-worked products use for subsequent processing techniques such as cold forming or for machining.
 - Sheet and strip fabricated into products or cold rolled into thinner, stronger material even into foil.
 - Blooms and billets rolled into finished products, such as structural shapes or railroad rail, or processed into semi-finished shapes, such as bar, rod, tube, or pipe

ROLLING

- Hot Rolling :-

- is prominent among all manufacturing processes
- equipment and practices are sufficiently advanced
- is standardized
- produce uniform-quality products at relatively low cost
- products are normally obtained in standard shapes and sizes

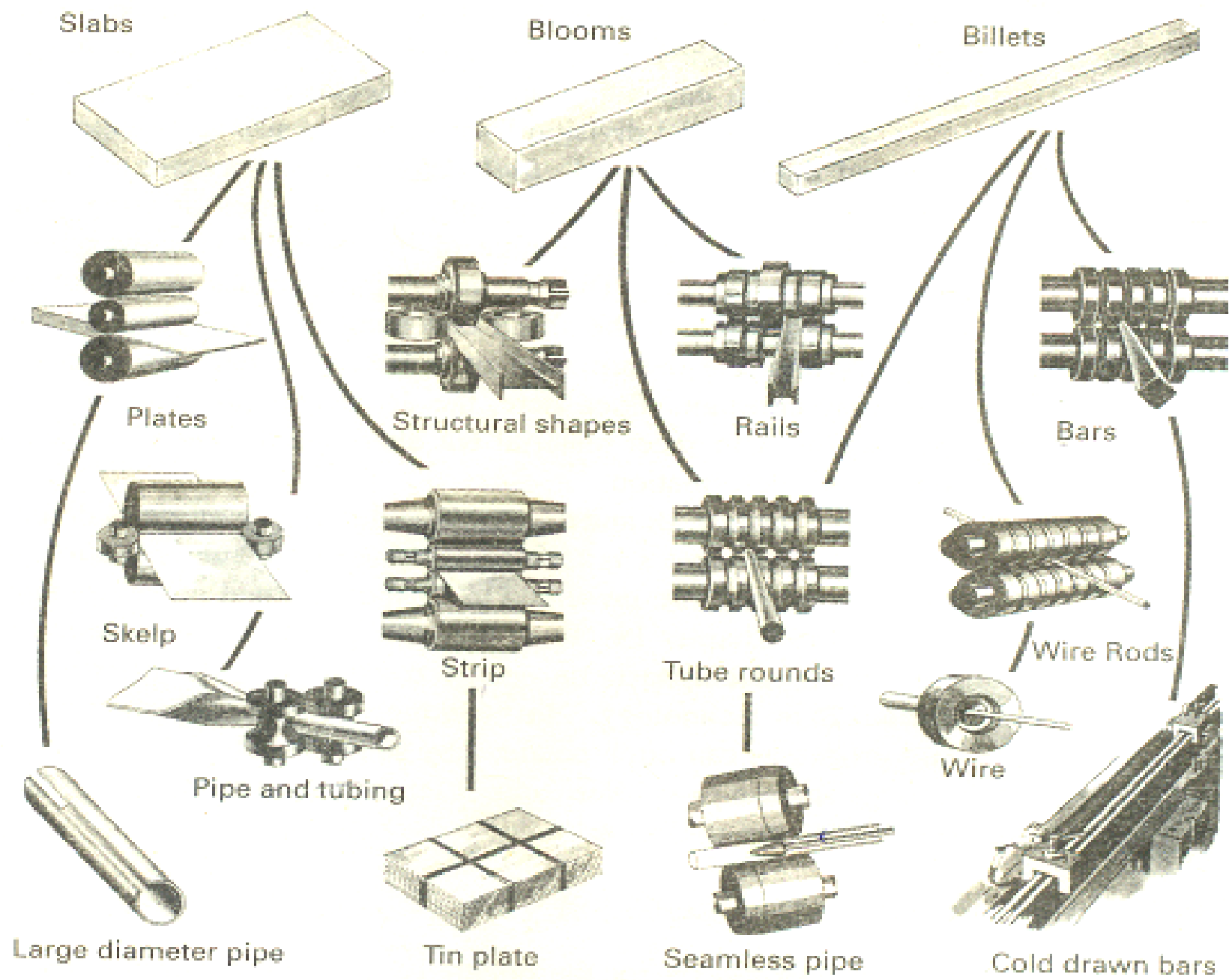


FIGURE 18-1 Schematic flowchart for the production of various finished and semifinished steel shapes. Note the abundance of rolling operations. (Courtesy of American Iron and Steel Institute, Washington, D.C.)

Basic Rolling Process

- Heated metal is passed between two rolls that rotate in opposite directions
- Gap between rolls is less than thickness of entering metal.
- Rolls rotate with surface velocity that exceeds speed of incoming metal, friction along the contact interface acts to propel the metal forward.

Basic Rolling Process

- Metal is squeezed and elongates result in decrease of the cross-sectional area.
- Amount of deformation in a single pass depends on the friction conditions along the interface.
- If too much material flow is demanded, rolls cannot advance the material and simply skid over its surface.
- Too little deformation per pass results in excessive production cost ??

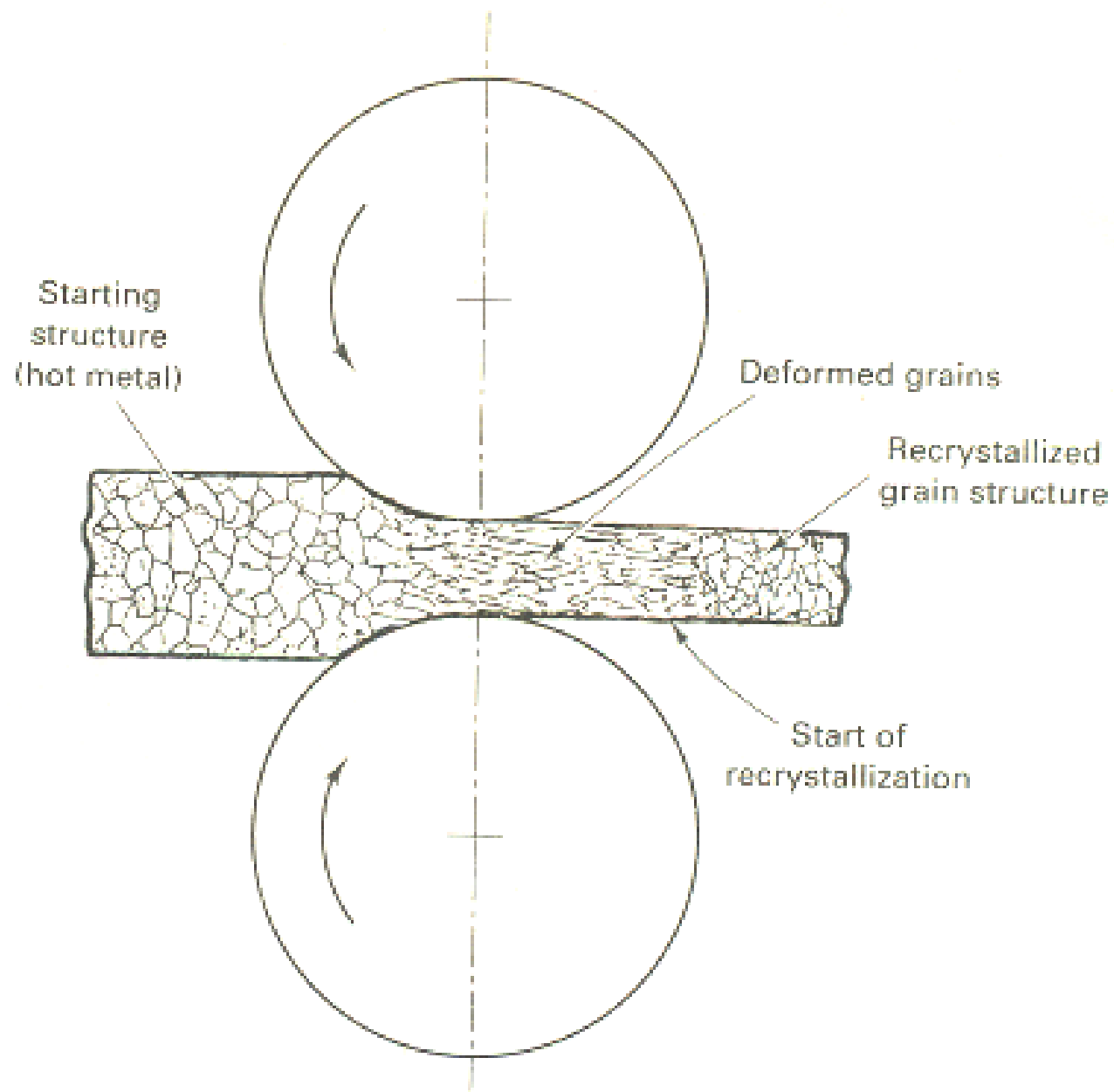


FIGURE 18-2 Schematic representation of the hot-rolling process, showing the deformation and recrystallization of the metal being rolled.

Rolling Temperatures

- Temperature control is crucial to the success of the hot rolling process.
- If the temperature of the billet is not uniform, the subsequent deformation will not be uniform.

Rolling Temperatures

- For example

- If a part cools prior to working, the cooler surfaces will tend to resist deformation.
- Cracking and tearing of surface may result as hotter, weaker interior tries to deform.

Rolling Temperatures

- Cooling from solidification is controlled to enable direct insertion into a hot-rolling operation without additional handling or reheating.
- Brought to rolling temperature, usually gas- or oil-fired soaking pits or furnaces are normally used.
- Plain-carbon and low-alloy steels soaking temperature is approximately 2200°F (1200°C)

Rolling Temperatures

- For smaller cross sections, induction coils may be used to heat material for rolling
- Hot rolling is usually terminated when temperature falls to about 100 to 200°F (50 to 100°C) above the recrystallization temperature of material
- Finishing temperature assures the production of a uniform fine grain size and prevents possibility of unwanted hardening
- Before additional deformation, a period of reheating is required to reestablish desirable hot-working conditions

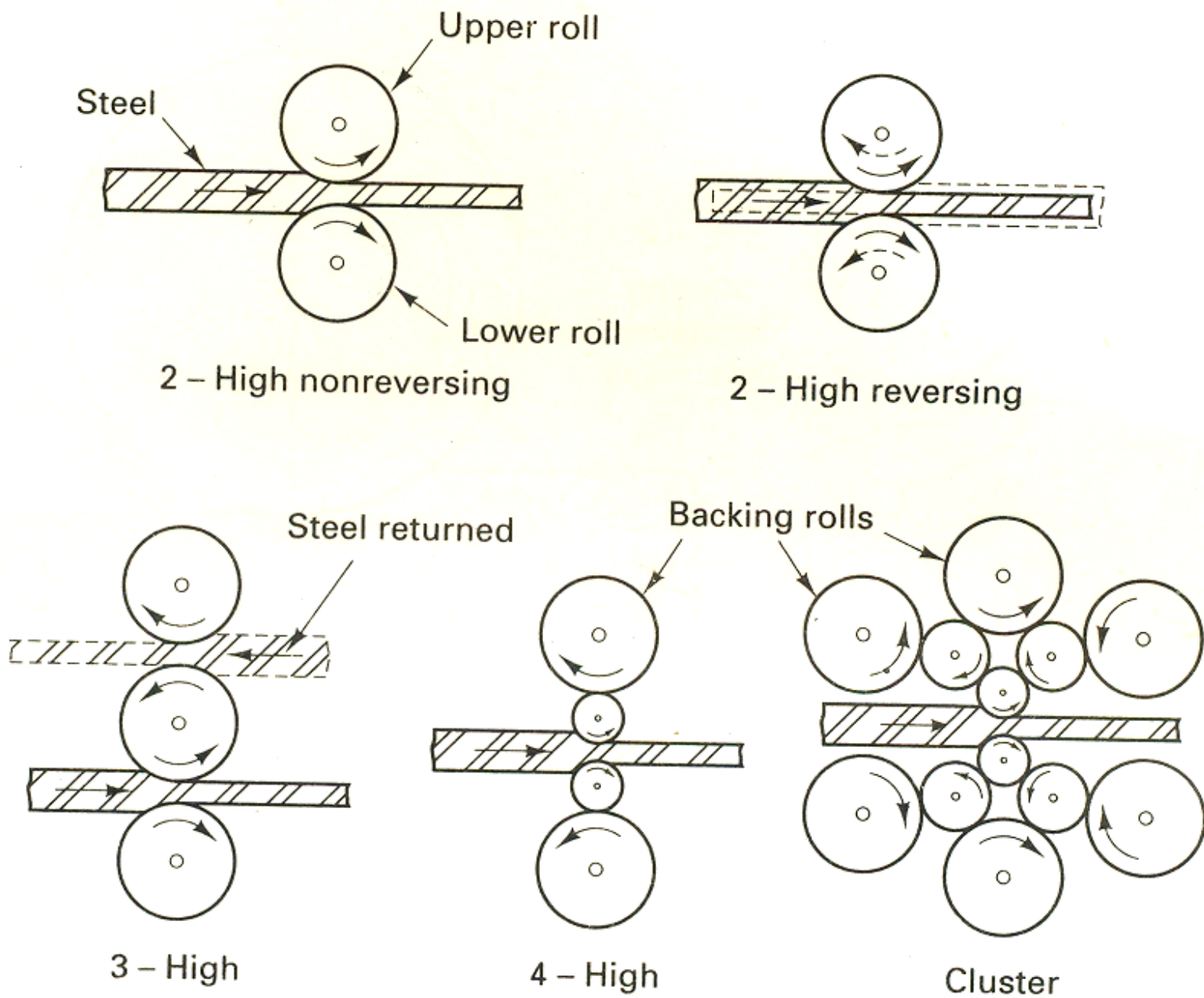


FIGURE 18-3 Various roll configurations used in rolling operations.

Rolling Mill Configurations

- Rolling mill stands are available in a variety of roll configurations.
- Early reductions (often called primary roughing or breakdown passes), employ two- or three-high configuration with 24- to 55-in. (600- to 1400-mm) diameter rolls
- Two-high non-reversing mill simplest design from which material can only pass in one direction
- Two-high reversing mill permits back-and-forth rolling, rolls may stop, reversed, and brought back to rolling speed between each pass

Rolling Mill Configurations

- The three-high mill eliminates need for roll reversal but requires some form of elevator on each side of mill to raise or lower material and mechanical manipulators to turn or shift product between passes
 - smaller-diameter rolls produce less length of contact for a given reduction and therefore require lower force and less energy to produce a given change in shape
 - smaller cross section, however, provides reduced stiffness and pressed apart by the metal passing through the middle

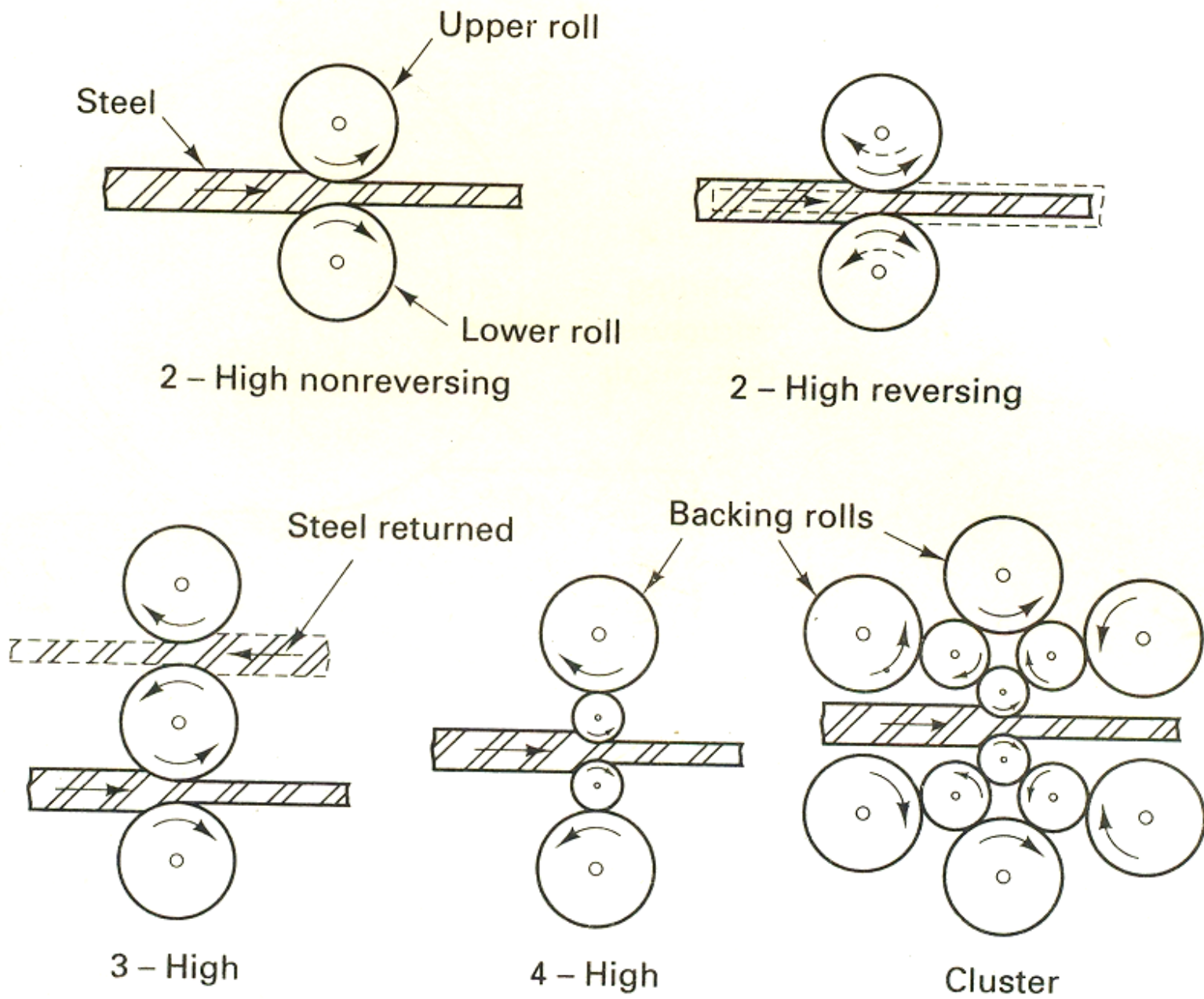


FIGURE 18-3 Various roll configurations used in rolling operations.

Rolling Mill Configurations

- Four-high and cluster arrangements use backup rolls to support the smaller work rolls
 - used in hot rolling of wide plate and sheets, and in cold rolling, where small negligence would result in an unacceptable variation in product thickness
 - Foil is rolled on *cluster mills* since small thickness requires small-diameter rolls
 - In a cluster mill, the roll in contact with the work can be as small as 1/4 in. in diameter

Rolling Mill Configurations

- Pack Rolling, a process where two or more layers of metal are rolled simultaneously as a means of providing a thicker input material
- Household aluminum foil is usually pack rolled, as evidenced by the one shiny side (in contact with the roll) and one dull side (in contact with the other piece of foil)
- In rolling of non-flat or shaped products, such as structural shapes and railroad rail, the sets of rolls contain contoured grooves that sequentially form desired shape, cross section and control metal flow

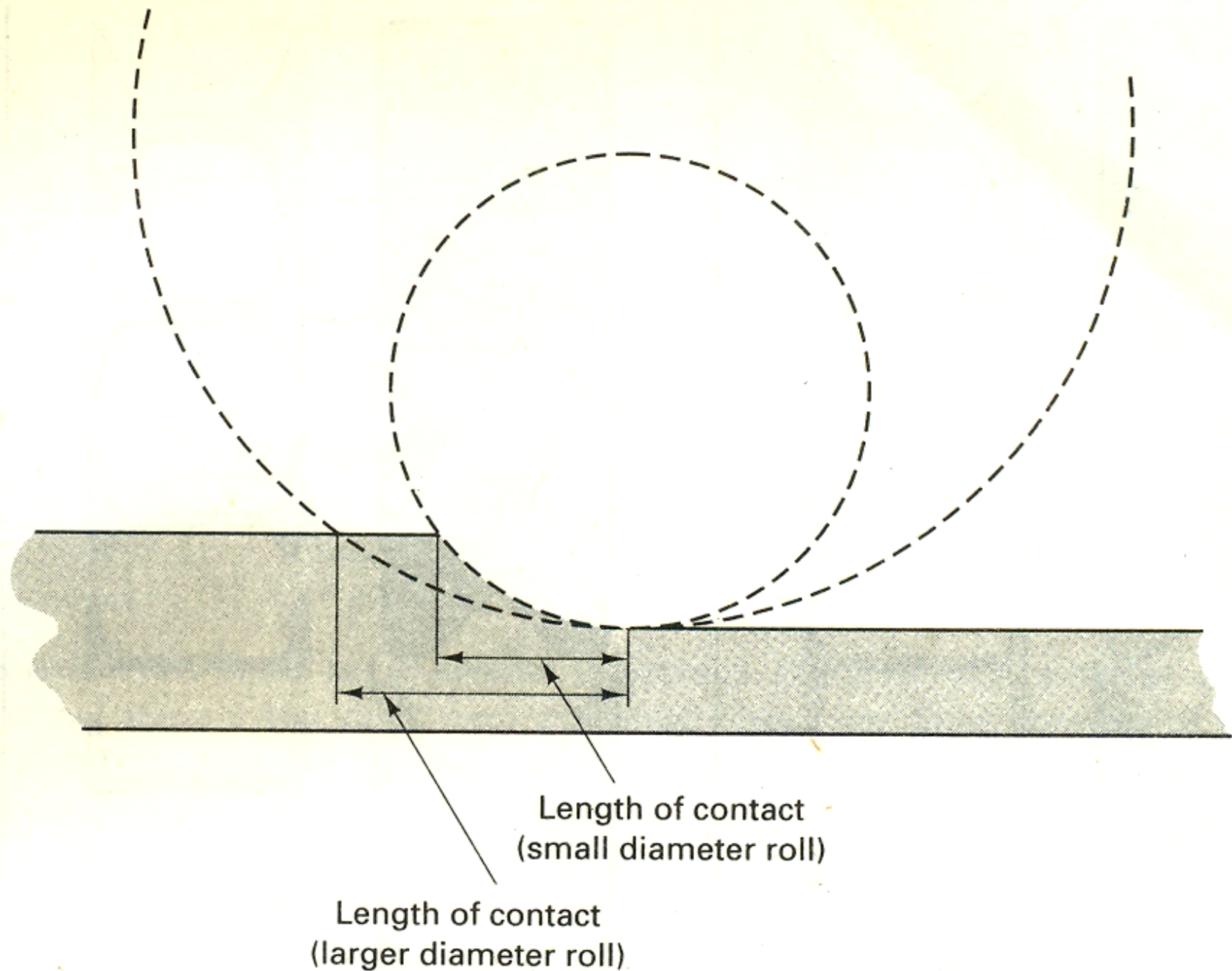


FIGURE 18-4 Schematic showing the effect of roll diameter on length of contact for a given reduction.

Continuous Rolling Mills

- When the volume of a product justifies investment, it may be rolled on a continuous rolling mill.
 - Billets, blooms, or slabs are heated and fed through an integrated series of non-reversing stands
 - Continuous mills for the hot rolling of steel strip, for example, often consist of a roughing train of approximately four four-high mill stands and a finishing train of six or seven additional four-high stands.

Continuous Rolling Mills

- In a continuous structural mill, the rolls in each stand contain only one set of shaped grooves, in contrast to the multi-grooved rolls used when the product is produced by back-and-forth passes through a single stand.
- In a continuous rolling mill, same amount of material must pass through each stand in a given period of time.
- If cross section is reduced, speed must be increased proportionately.

Continuous Rolling Mills

- Thus rolls of each successive stand must turn faster than those of preceding one by an amount equal to change in cross-sectional area
- If this synchronization is not maintained, material may accumulate between stands, or demand for incoming material may place material under excessive tension, and cause a tearing or rupture

Continuous Rolling Mills

- Synchronization of six or seven mill stands is not an easy task, especially when key variables such as temperature and lubrication may change during a single run and product may be exiting final stand at speeds in excess of 70 miles per hour (110 kilometers per hour).
- Computer control is important to successful rolling, and modern mills are equipped with numerous sensors to provide the needed information.

Ring Rolling

- In ring rolling process, one roll is placed through the hole of a thick-walled ring, and a second roll presses in from outside.
- As the rolls squeeze and rotate, wall thickness is reduced and diameter of ring increases.
- Shaped rolls can be used to produce a wide variety of cross-section profiles.
- Resulting seamless rings find application in products such as rockets, turbines, airplanes, pipelines, and pressure vessels.

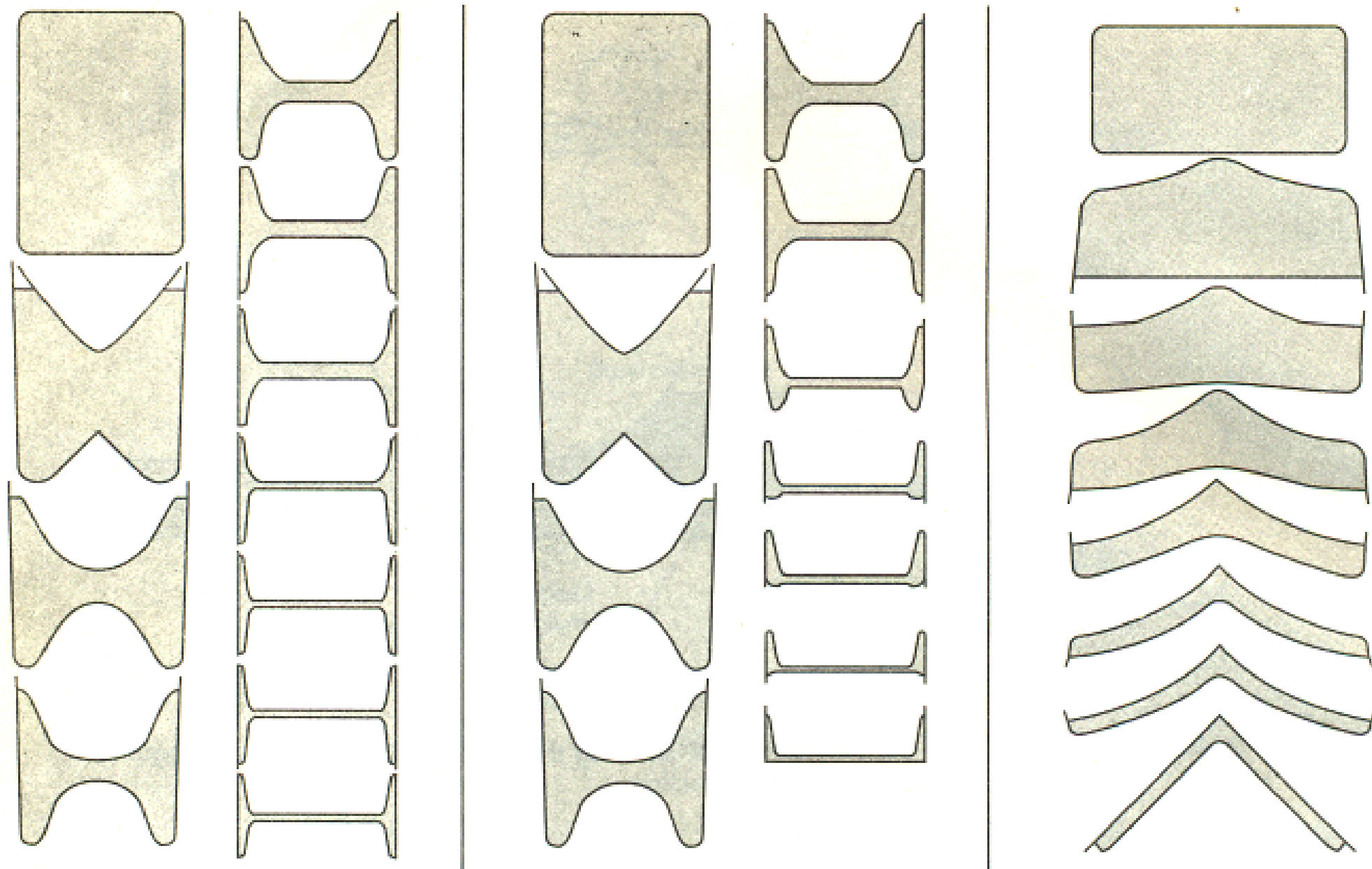


FIGURE 18-5 Typical roll-pass sequences used in producing various structural shapes.

Characteristics of Hot-Rolled Products

- Because they are rolled and finished above recrystallization temperature, hot-rolled products have little directionality in their properties and are relatively free of deformation - induced residual stresses.
- These characteristics may vary, depending on thickness of product and presence of complex sections.

Characteristics of Hot-Rolled Products

- Substantial residual stresses can be induced during hot working.

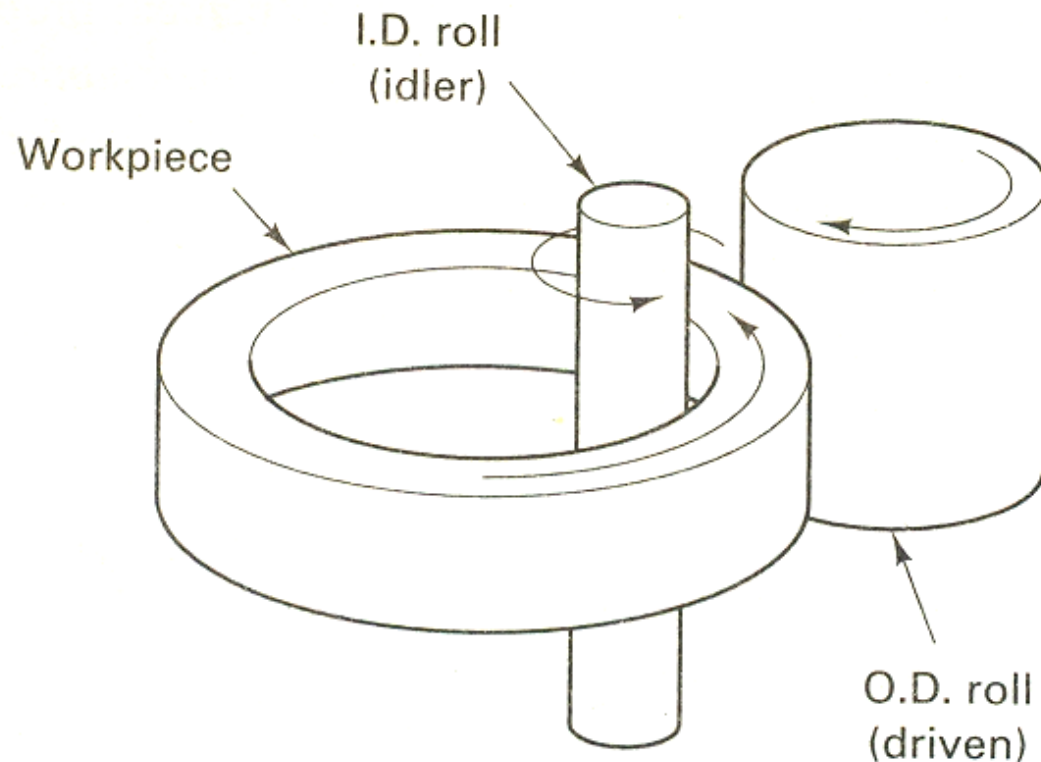


FIGURE 18-6 Schematic of a horizontal ring rolling operation. As the thickness of the ring is reduced, its diameter will increase.

Characteristics of Hot-Rolled Products

- Thin sheets often show some definite directional characteristics, whereas thicker plate (such as that above 0.8 in. or 20 mm) will usually have very little.
- Because of the high residual stresses in rapidly cooled edges, a complex shape, such as an T- or H-beam, may warp noticeably if a portion of one flange is cut away.

Characteristics of Hot-Rolled Products

- As result of hot deformation and good control hot-rolled products are normally of uniform and dependable quality and reliability.
- It is quite unusual to find any voids, seams, or laminations when these products are produced by reliable manufacturers.
- Surfaces of hot-rolled products are usually a bit rough and are originally covered with a tenacious high-temperature oxide, known as mill scale.

Characteristics of Hot-Rolled Products

- Removed by an acid pickling operation, resulting in a surprisingly smooth surface finish.
- Dimensional tolerances of hot-rolled products vary with kind of metal and size of the product.
- For most products produced in reasonably large tonnages, tolerances are within 2 to 5% of specified dimension (either height or width).

Flatness Control and Rolling Defects

- Rolling of flat material with uniform thickness requires uniform gap between rolls attaining such an objective may be difficult.
- Consider upper roll in a set that is rolling sheet or plate material presses upward in the middle of roll supported in mill frame.

Flatness Control and Rolling Defects

- Roll is loaded in three-point bending and tends to flex in a manner that produces a thicker center and thinner edge.
- If roll is always used to roll same material at same temperature, forces and deflections can be predicted, and roll can be designed to have a specified amount of crowning.
- When roll is subjected to a specified load, it will “deflect into flatness”.

Flatness Control and Rolling Defects

- If applied load is not of designed magnitude, profile will not be flat and defects may result.

The thinner material will try to become longer but must remain attached to the thicker. Result may be wavy edges or fractures in center. If correction is excessive, center becomes thinner and longer, and result can be a wavy center or cracking of the edges.

Thermo-mechanical Processing and Controlled Rolling

- A rolling process is generally used as being a means of changing shape of material.
- Heat may be used to reduce forces and promote plasticity while mechanical properties (heat treatments) are usually performed as subsequent operations.
- Thermo-mechanical processing consists of both deformation and controlled thermal processing to produce desired levels of strength and toughness in the working product.

Thermo-mechanical Processing and Controlled Rolling

- Possible goals of thermo mechanical includes :-
 - Production of uniform fine grain size
 - Controlling nature
 - Size and distribution of various transformation products (such as ferrite, pearlite, bainite, and martensite in steels)
 - Controlling the reactions that produce solid solution strengthening or precipitation hardening
 - Producing a desired level of toughness.

Thermo-mechanical Processing and Controlled Rolling

- Following must all be specified and controlled:-
 - Starting structure (controlled by composition and prior thermal treatments),
 - deformation details,
 - temperature during the various stages of deformation,
 - the cool down from the working temperature.

Thermo-mechanical Processing and Controlled Rolling

- Computer-controlled rolling facilities are almost a necessity if thermo-mechanical processing is to be performed successfully.
- Possible benefits of thermo-mechanical processing include
 - improved product properties;
 - substantial energy savings (by eliminating subsequent heat treatment);
- Possible substitution of a cheaper, less-alloyed metal for a highly alloyed one that responds to heat treatment.